IM-CMDet: An Intramodal Enhancement and Cross-Modal Fusion Network for Small Object Detection in UAV Aerial Visible-Infrared Imagery

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***Abstract*—Uncrewed aerial vehicle (UAV) aerial visible–infrared [RGB-thermal (RGBT)] object detection has been widely applied in fields such as military operations and rescue missions. However, although numerous UAV aerial RGBT object detection methods exist, several challenges remain in this field. On the one hand, drones typically operate at high altitudes, and objects only occupy a small number of pixels in imaging, posing a significant challenge to object detection. On the other hand, spatial misalignment between modalities remains a major obstacle in cross-modal fusion—especially given the small size of the objects. To address the above issues, this article proposes IM-CMDet, an intramodal enhancement and cross-modal fusion network for small object detection in UAV-based RGBT imagery, which comprises three effective modules: the detail-semantics joint enhancement (DSJE) module, the differential-based fusion weight generation (DFWG) module, and the feature reconstruction network (FRN). The DSJE module prevents small object features from being overwhelmed by background noise through optimizing feature representations across different levels. The FRN module is designed to overcome modality differences and build intermodality information correlation via the Swin-Transformer architecture. To further enhance the network's sensitivity to small objects, the DFWG combines differential and spatial attention to generate the final fusion weights while reducing the impact of background noise on detection performance. Extensive experiments on RGBTDronePerson and two additional benchmarks demonstrate that IM-CMDet achieves state-of-the-art performance through effective cross-modal fusion, significantly advancing small-object detection in complex aerial scenarios. The code is available at https://github.com/RS-Minchao/IM-CMDet**

***Index Terms*—Aerial images, multimodal fusion, small object detection, thermal images.**

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NOMENCLATURE

CBR	Convolution-batchnorm-rectified linear unit (ReLU) block.
Conv(A, B)	Convolution operation with kernel B on input A.
$\odot$	Elementwise Hadamard product.
$\cdot$	Broadcast-based elementwise product.
GAP	Global average pooling.
Nearest$(\cdot)$	Nearest-neighbor interpolation.
SPA	Spatial attention mechanism.
Linear$(\cdot)$	Linear projection operation.
I	Input image.
F, F_V, and F_I	Feature pyramids extracted from input, where $F \in \{F_V, F_I\}$.
f_V and f_I	Feature layers from F_V and F_I, respectively.
F^D, F_V^D, and F_I^D	Output pyramids from DSJE module, where $F^D \in \{F_V^D, F_I^D\}$.
f_V^D and f_I^D	Feature layers from F_V^D and F_I^D.

I. INTRODUCTION

UNMANNED aerial vehicles (UAVs) have become an important tool for rapidly acquiring ground object information due to their excellent maneuverability, wide imaging range, and low cost. They are widely used in fields such as agricultural management [1], search and rescue [2], and traffic monitoring [3]. With the development of neural networks, the performance of methods in the field of computer vision has significantly improved [4], [5], and similarly, numerous methods have emerged in UAV-based object detection [6], [7], [8]. However, despite the increasingly widespread applications of UAVs, several challenges remain, such as high-altitude imaging environments and low-light conditions. Consequently, object detection in UAV-based images continues to be an active yet challenging research task.

Conventional object detection methods are primarily designed for natural images and often fail to account for the significantly reduced size of objects in UAV-captured imagery due to high-altitude imaging conditions, leading to limited detection performance. In reality, objects in UAV images exhibit much smaller pixel dimensions compared to natural images. For categories that are inherently small in ground-level images (such as pedestrians and bicycles), their pixel areas

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in UAV images may account for less than 0.1% of the entire image, which significantly increases the difficulty of detection. In the field of small object detection, several studies have addressed these challenges [9], [10], [11], [12]. For instance, Bosquet et al. [13] proposed a generative adversarial network (GAN)-based data augmentation method to mitigate the scarcity of small training samples, thereby enhancing model performance while reducing the reliance on extensive manual annotations. Zhang et al. [14] introduced a random-connection attention (RCA) module that integrates convolutional operations with attention mechanisms to jointly extract global and local features, adaptively focusing on sparse correlated regions while suppressing irrelevant background pixels. Jing et al. [15] and Ma et al. [16] improved small object detection in low-resolution UAV images through hierarchical feature fusion and a visual perception module. To address performance redundancy in existing small object extraction methods, Zhang et al. [17] further developed a wavelet-regularized channel pruning strategy, maintaining accuracy while reducing computational complexity. However, on the one hand, the aforementioned methods are insufficient in preserving information of small objects, leading to the gradual loss of small object signals as the network deepens. On the other hand, traditional object detection methods, which mostly rely on single-modality visible images, face many challenges. The imaging quality of visible images is significantly impacted by factors such as lighting variations and weather conditions, resulting in notable limitations in detection accuracy and applicability in low-visibility environments, such as rain, fog, and nighttime conditions [18]. As a result, the robustness and reliability of these single-modality methods are greatly insufficient. In contrast, thermal images have a natural advantage in low-light and adverse weather conditions [19], as they can clearly present the position and contours of objects through thermal radiation differences [9], [10]. However, thermal images lack color and detail information, unable to provide the same rich visual content as visible images. Therefore, cross-modal object detection methods that fuse visible and thermal images have gradually become a research hotspot and are widely applied in fields such as environmental monitoring [20], vehicle surveillance [21], and crowd density estimation [22]. These methods leverage the complementary advantages of both modalities to provide more robust and accurate object detection capabilities in various environments.

Based on the fusion stages of visible and thermal image fusion for object detection, related methods can be classified into pixel-level fusion [23], feature-level fusion [24], and decision-level fusion [25]. Among these fusion methods, pixel-level fusion occurs at the image layer and aims to achieve high-precision image enhancement for improved detection accuracy. However, these methods primarily focus on pixel-level relationships between images and seldom consider object semantics. When combined with object detection tasks, the fusion effect is typically constrained by detection loss, but the effectiveness of this approach remains limited [23]. Decision-level fusion uses probability distribution networks to combine multiple independent detectors and derive the final detection results by integrating their outputs. However, decision-level methods require datasets with annotations in both modalities, which poses a challenge for their application. In contrast, feature-level fusion methods use cross-modal feature fusion strategies to explore the complex correlations between modalities, thereby achieving better performance [26], [27]. As a result, feature-level fusion methods have gradually become the mainstream research approach. Based on the fusion methods, they can further be divided into two categories: static fusion and adaptive fusion methods.

Static fusion methods, which are more commonly used in the field, achieve effective feature fusion and enhance the network's detection performance by integrating multimodal features with reference to corresponding pixel locations through predefined architectures. For instance, Liu et al. [28] propose an object-aware adversarial learning network that learns differences while preserving commonalities between various modal features. Cao et al. [29] introduce an image observation-based pedestrian detection method that utilizes essential local information from reference modes to guide cross-modal feature aggregation. Cao et al. [30] proposed a novel lightweight fusion module that effectively integrates inputs from different modalities using channel switching and spatial attention. Zhang et al. [31], on the other hand, leverage illumination information from images to guide cross-modal feature weighting for better utilization of complementary information. However, these methods are designed for natural images without considering the unique characteristics of UAV imagery, leading to limited detection performance. Recently, UAV-based optical-infrared joint object detection technology has garnered significant attention, with numerous researchers dedicated to related endeavors [32], [33]. For aerial image object detection purposes, a cross-modal attention feature fusion module is proposed that can effectively learn shared features as well as distinguish different features within each mode in [34]. Sun et al. [26] devised an uncertainty sensing module that employs cross-modal intersections and lighting assessments to quantify the uncertainty of each object, effectively leveraging information from both modes for improved object detection outcomes. However, this method requires comprehensive annotations for both modalities, resulting in a significant annotation burden and limiting the method's adaptability. Based on an analysis of the visible and infrared spectral ranges, Jiang et al. [35] innovatively combined two out of the three RGB channels with the infrared band for feature extraction. Wang et al. [36] and Zhao et al. [37] designed a three-branch feature enhancement network to enhance the learning of features from each modality and the fused modality, introducing cross-modal attention feature fusion to fully extract shared and complementary features between the two modalities. The aforementioned method fails to account for the fact that aerial imaging results in objects occupying only a small number of pixels, making them susceptible to being overwhelmed by background features. Consequently, its performance in small object detection is constrained. To address this, Zhang et al. [38] proposed a quality-aware learning strategy to address the issue of data quality imbalance from a fine-grained perspective, thereby improving the quality of modality fusion and enhancing detection performance. Regrettably, these static fusion methods fail to consider the spatial misalignment of objects between modalities during

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direct fusion, hindering the network's ability to effectively exploit the complementary information between modalities and ultimately degrading object detection performance [39].

To address the limitations of the aforementioned static fusion method Training and Testing Details, some researchers have focused on adaptive fusion, designing networks that adapt to the characteristics between different data sources in order to alleviate the weak alignment of object positions between modalities [40], [41]. To address the issues of modality calibration errors and inaccurate fusion, a new calibration complementary Transformer called C2former is proposed in [42], leveraging the powerful feature correlation learning capabilities of Transformers [43]. Despite yielding promising results, this component still imposes significant computational overhead due to the extensive use of attention mechanisms. To address the issue of limited detection performance caused by the misalignment of image pairs obtained from real-world scenarios, Song et al. [44] proposed a strategy called object search rectification, which performs registration prior to modality fusion. Although the aforementioned methods aim to reduce the modality gap through alignment operations, they place excessive emphasis on strict alignment requirements, which makes detection performance heavily dependent on the quality of the alignment. To address this, Chen et al. [45] proposed a novel method called offset-guided adaptive feature alignment, combining offset-based reference generation and deformable convolution [46]. Liu et al. [47] proposed the cross-mamba interaction and offset-guided fusion for multimodal object detection (COMO) framework, which leverages cross-Mamba interaction and offset-guided fusion to achieve efficient multimodal feature alignment and interaction for improved object detection. To address challenges from modality inconsistency and redundancy, Bao et al. [48] designed a dual-dynamic cross-modal interaction network that employs channel-gated spatial cross-attention and dynamic feature fusion. By combining deformable convolution with cross-modal global context modeling (CGC) (derived from GCNet [49]), local correction for both modes and global context learning is achieved to enhance network performance in [50]. Wang et al. [51] proposed a novel cross-scale dynamic convolution fusion module to adaptively extract and fuse bimodal features with respect to data distribution, thereby enhancing the detection performance of the method. These approaches enable the network to implicitly capture the optimal fusion locations for detection tasks, rather than relying on strict alignment. However, during high-altitude UAV operations, small objects like pedestrians and bicycles occupy less than 0.1% of total pixels, resulting in cross-modal misalignment that can reach several times the objects' own scales, significantly increasing the difficulty of feature matching. Unfortunately, the aforementioned methods fail to account for this issue, resulting in insufficient attention to small object features and the absence of a dedicated fusion strategy, ultimately limiting their performance in small object detection.

Based on the analysis of the aforementioned methods, current UAV-based RGB-thermal (RGBT) fusion methods for small object detection have several shortcomings that limit their effectiveness, which can be identified as follows.

Issue 1: For UAV-based RGBT object detection methods, the design does not take into account the characteristics of high-altitude imaging. Certain object categories, such as pedestrians and bicycles, occupy less than one-thousandth of the entire image, causing these discriminative features to be easily overwhelmed by background information during the network learning process.

Issue 2: When the object size is small, the issue of cross-modal misalignment becomes more pronounced. However, few fusion methods are specifically designed for small object detection, which may hinder the effective integration of complementary features across modalities.

To address the aforementioned issues, this article proposes a small object detection algorithm based on the fusion of UAV aerial RGBT images, termed IM-CMDet. Given a pair of visible and infrared images, initial features of each modality are independently extracted by their corresponding backbone networks. These features are, then, input into the detail-semantics joint enhancement (DSJE) module, which is devised from the feature pyramid network (FPN) [52]. In this module, high-frequency details extracted from the original images, along with high-level semantic features, are used to enhance the features at each level, enabling the long-term retention of important discriminative information. To further guide the network's attention to the hierarchical and channelwise differences in object features, DSJE introduces a new hierarchical channel attention mechanism that adjusts feature weights based on both hierarchical and channel characteristics. The enhanced features are subsequently fed into the feature reconstruction network (FRN) with an asymmetric cross-attention mechanism, specifically designed to mitigate significant misalignment in small objects, effectively reconstructing visible modality information based on infrared modality information. Finally, fusion weights, generated by combining the differential process and spatial attention from the initial features, are applied to the weighted fusion of visible and infrared modalities, reducing the impact of redundant information on detection performance while maintaining sensitivity to small object locations. This approach significantly enhances the network's performance in small object detection.

The main contributions of this article are as follows.

1) We propose a novel RGBT fusion network to enhance UAV-based small object detection under varying illumination conditions. By integrating discriminative feature enhancement with a cross-modal feature alignment network specifically designed for small object characteristics, our method effectively mitigates the limitations of existing small object detection approaches in this domain, leading to a significant improvement in UAV-based RGBT fusion detection performance.
2) To address the issue of small object features being easily overwhelmed by background information during the learning process, we design the DSJE module, which integrates semantic features with high-frequency information from the original image to enhance discriminative feature representation. Furthermore, a hierarchical channel joint attention mechanism is introduced to adaptively adjust feature representations at different levels.

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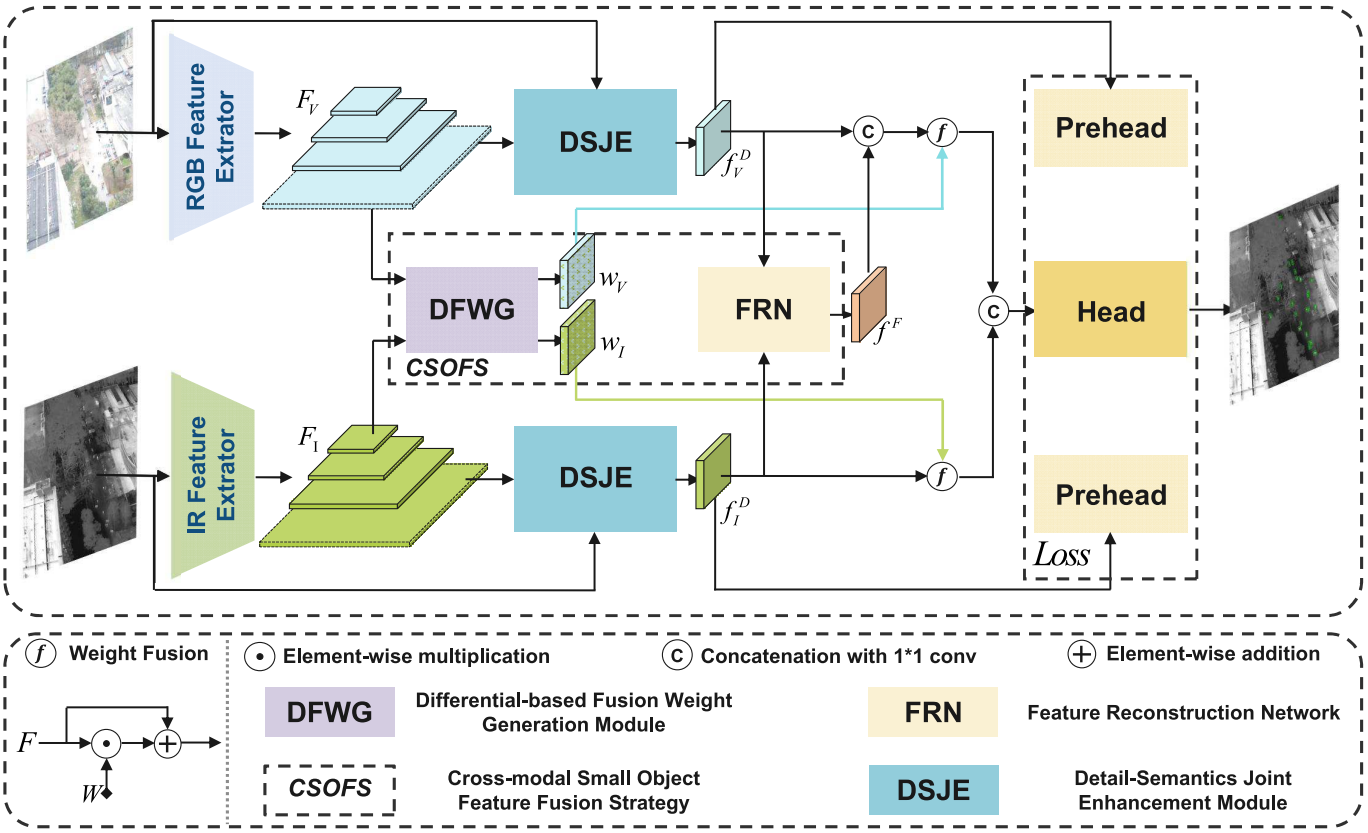


Fig. 1. Overall framework of IM-CMDet.

3) We propose a fusion strategy for cross-modal small object detection, combining differential-based fusion weight generation (DFWG) with FRN. The former enhances sensitivity to small objects and reduces redundancy, while the latter models intermodal correlations, enabling effective fusion.

The pipeline of the remaining sections of this article is as follows. In Section II, we detail the architecture of the network. In Section III, we present this work, including the experimental results and relevant comparisons to validate the effectiveness of our approach. Finally, in Section IV, we summarize and discuss this work.

II. Methodology

A. Overview

The overall framework of the proposed IM-CMDet is illustrated in Fig. 1. The visible and thermal modal features are extracted using a two-stream backbone network. Subsequently, the extracted features undergo processing via the DFWG to produce the necessary fusion weights for the final integration. These features are, then, fed into our proposed DSJE, which refines the extracted features from both semantic and detailed perspectives, thereby enhancing the salience of regions of interest and complementing the shallow texture features of the object. During the training phase, the enhanced features are input into the predetection head, where loss calculation optimizes the feature extraction capabilities of the preceding network stages. FRN with the asymmetric cross-window attention mechanism based on thermal modalities is subsequently employed to augment the feature representation of visible modalities and establish correlations between the two modal objects. Finally, after adjusting the fusion weights, the weighted fusion features are input into the detection head to predict object-related information within the image.

B. DSJE Module

As discussed, the discriminative features of small objects tend to be gradually diminished or ultimately overwhelmed by cluttered background contexts during iterative learning procedures. To address this issue, we have developed the DSJE module. This module extracts detailed features from original images and high-level semantic features, thereby enhancing the salience and representativeness of object features from both detailed and semantic perspectives. An overview of the DSJE module is provided in Fig. 2. To ensure model efficiency, we use the C3–C5 layers from the backbone network as inputs.

We obtain the initial detail information H of the image using the Laplacian operator and then generate the detail feature map through convolution operations on this initial detail information

$$\mathrm{LAP} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \tag{1}$$

$$H = \mathrm{CBR}(I, \mathrm{LAP}) = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(I, \mathrm{LAP}))) \tag{2}$$

where the I represents the initial visible or thermal image, the CBR consists of convolution, batch normalization, and

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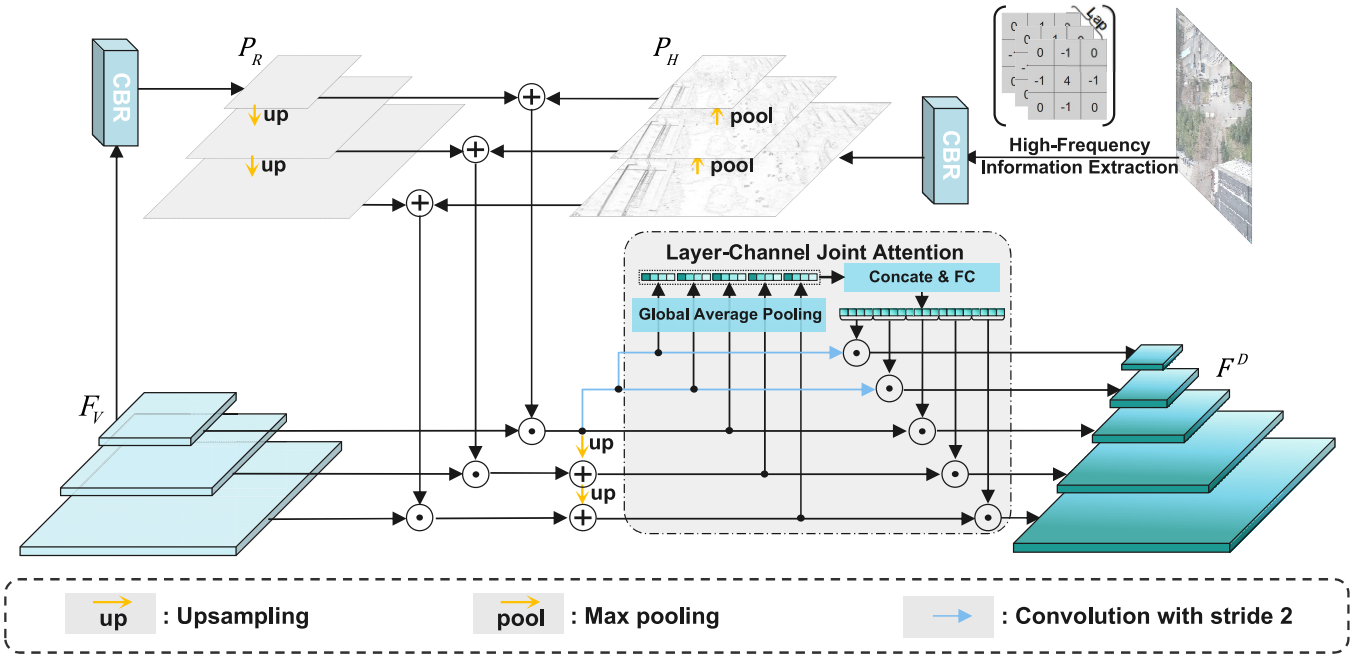


Fig. 2. Framework of DSJE module.

ReLU, and Conv(A, B) denotes the convolution of input A with kernel B.

Given the extensive coverage of UAV imagery and the high density of ground objects, edge detail features extracted using the Laplacian operator frequently contain significant noise. To mitigate the adverse effects of this noise on subsequent detection accuracy and to enhance the feature representation of regions of interest in shallow layers, this method leverages deep semantic features to generate enhanced References (ERs). Specifically, the final ER R is produced by applying convolutional operations and threshold segmentation to the deepest feature maps of the input data

$$R(x,y) = \begin{cases} K, & \text{if } \mathrm{CBR}(f_3)(x,y) \geq T \\ 1, & \text{if } \mathrm{CBR}(f_3)(x,y) < T \end{cases} \tag{3}$$

where f_3 refers to the deepest semantic features of the input feature pyramid F, (x, y) represents the pixel location, and K is an empirical value set to 4 here.

We subsequently upsample the ER features while downsampling the detail features to construct a feature pyramid corresponding to each layer's scale. We, then, integrate the detail feature pyramid P_H with the ER pyramid P_R, thereby generating the final detail semantic enhancement weights W_D

$$P_R[i] = \mathrm{Nearest}\left(R, 2^{(3-i)}\right), \quad i = 1, 2 \tag{4}$$

$$P_H[i] = \mathrm{MaxPool}\left(H, 2^i\right), \quad i = 1, 2 \tag{5}$$

$$W_D = P_R + P_H \tag{6}$$

where the Nearest$(\cdot)$ represents the nearest neighbor interpolation operation, the MaxPool$(\cdot)$ represents the maxpooling operation, and the left side of the parentheses represents the original feature map for the operation, while the right side indicates the multiplicative factor for the operation.

Subsequently, multiply the enhanced weights by the original features of each layer to obtain the refined features F_e. Analogous to the FPN, these refined features facilitate information exchange between adjacent layers. To accommodate potentially large-scale objects, we further apply convolutional layers on the top-level features of F_C to extract deeper semantic representations

$$F_e = F \odot W \tag{7}$$

$$F_C[i] = \begin{cases} F_e[i], & \text{if } i = 3 \\ F_e[i] + \mathrm{Nearest}\left(F_C[i+1], 2\right), & \text{if } i = 2, 1 \end{cases} \tag{8}$$

$$F_C[i] = \mathrm{Conv}_{s=2}\left(F_C[i-1]\right), \quad i = 4, 5 \tag{9}$$

where $\odot$ denotes the Hadamard product, F_C represents the feature after interaction, the values of i are selected in the descending order in (8), and $\mathrm{Conv}_{s=2}$ represents a convolution operation with a stride of 2.

Driven by the need to clearly identify the task requirements of each channel and hierarchical level, we aim to autonomously enhance the feature information of a particular layer by comparing information across different levels. Consequently, we propose a hierarchy-channel joint attention mechanism. Analogous to channel attention mechanisms, we derive hierarchical channel weights by applying GAP to the features of each channel at various levels. Subsequently, the features from each level are aggregated and processed through a fully connected layer to generate new hierarchical channel weights. These weights are, then, multiplied with the original features to obtain the refined final features F^D

$$l_i = \mathrm{GAP}\left(F_C[i]\right), \quad i = 1, 2, 3, 4, 5 \tag{10}$$

$$F^D = F_C \cdot \mathrm{Linear}\left(\mathrm{Cat}\left(l_1, l_2, l_3, l_4, l_5\right)\right) \tag{11}$$

where GAP denotes the global average pooling, $\cdot$ represents the multiplication based on the broadcasting mechanism, and Cat represents the concatenation operation.

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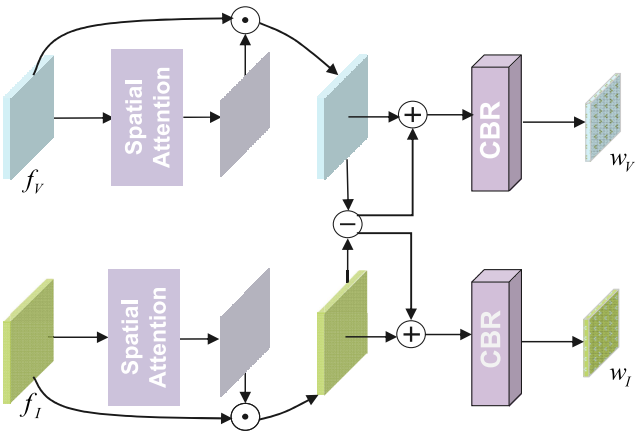


Fig. 3. Framework of DFWG module.

C. Cross-Modal Small Object Feature Fusion Strategy

Due to factors such as viewpoint deviation and resolution differences, there exists a positional discrepancy between objects in visible and thermal modes. Previous methods have seldom considered the size of the object when designing cross-modal feature interaction and fusion strategies. As the number of pixels occupied by the object decreases, the positional deviation becomes increasingly significant, leading to suboptimal fusion outcomes. In this article, we propose a specialized cross-modal small object feature fusion strategy (CSOFS). This strategy comprises two primary components: a difference-based fusion weight generator and an asymmetric window cross-attention-based cross-modal small-object feature alignment network.

1) DFWG Module: To mitigate the impact of redundant information from both modalities on the final detection accuracy, we generate fusion weights before fusion. This process aims to amplify the effective information of each modality while suppressing the expression of redundant data. Given the considerable positional deviation of small objects in UAV images across different modalities, we employ a differential approach that enables the network to emphasize object regions by leveraging significant positional deviations while simultaneously reducing background noise. This enhances the sensitivity and accuracy of the small object detection network. The detailed implementation is illustrated in Fig. 3.

The characteristics of the visible and thermal modalities are processed to generate a spatial attention mask. This mask is, then, applied to the original features through the Hadamard product, resulting in the reconstructed features. Subsequently, the reconstructed features from both modalities are combined using pixelwise subtraction to obtain the feature difference. By integrating these different results with the respective modality features, the final fusion weight $W_{V/I}$ can be derived. Notably, to fully leverage the representational power of shallow features for small object detection, we use the C2–C5 layers extracted from the backbone network as input. The formula for this process is presented as follows:

$$w_V = \mathrm{CBR}_{s=2}\left(2\left(f_V\right) \odot \mathrm{SPA}\left(f_V\right)\right) - f_I \odot \mathrm{SPA}\left(f_I\right) \tag{12}$$

$$w_I = \mathrm{CBR}_{s=2}\left(2\left(f_I\right) \odot \mathrm{SPA}\left(f_I\right)\right) - f_V \odot \mathrm{SPA}\left(f_V\right) \tag{13}$$

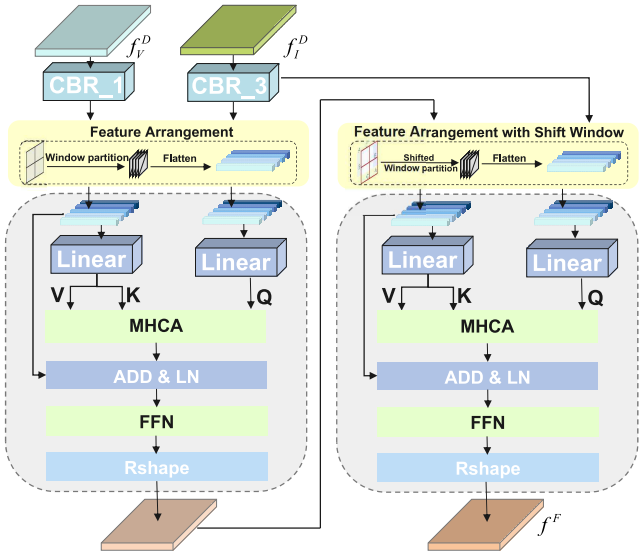


Fig. 4. Framework of FRN.

where $\mathrm{CBR}_{s=2}$ denotes a CBR module with a convolutional stride of 2, $\odot$ represents the Hadamard product, and SPA stands for spatial attention. The $f_{V/I}$ is derived from the original input feature pyramid $F_{V/I}$.

2) FRN Based on Small Object Correlation: Significant positional deviations of small objects across modalities must be addressed before fusion. Some previous methods have employed feature alignment operations; however, achieving both strict and efficient feature alignment has proven challenging. Consequently, some researchers have adopted deformable convolution to enable networks to autonomously learn correlations between specific locations and other regions, thereby facilitating feature reconstruction. However, training deformable convolution remains difficult, complicating the learning of precise position offsets. Since the introduction of Transformer architectures, the attention mechanism has gained widespread adoption in computer vision. In the context of cross-modal feature interaction, cross-attention mechanisms have demonstrated considerable potential. Given that thermal modality images exhibit stable quality unaffected by illumination conditions, this study leverages a cross-attention mechanism based on thermal features to enhance the representation of visible modality features. To reduce computational burden, this method employs cross-window attention. An overview of this approach is illustrated in Fig. 4.

Due to the significant positional discrepancy between visible and thermal modalities for small- and medium-sized objects in UAV imagery, a uniformly positioned window may encompass disparate objects following window segmentation. To mitigate the adverse effects of this issue on network performance, convolutional kernels of varying dimensions are employed to extract information from the enhanced features of each modality, as illustrated in Fig. 4

$$f_V^a = \mathrm{CBR}_{3\times3}\left(f_V^D\right) = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(f_V^D\right)\right)\right) \tag{14}$$

$$f_I^a = \mathrm{CBR}_{1\times1}\left(f_I^D\right) = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(f_I^D\right)\right)\right) \tag{15}$$

where the $\mathrm{CBR}_{3\times3}$ represents the CBR with 3×3 kernel size and the meaning of $\mathrm{CBR}_{1\times1}$ is similar.

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To facilitate the subsequent application of local cross-attention, we partition and flatten the obtained feature map. In line with the Swin-Transformer architecture [53], the left branch employs standard window partitions, while the right branch utilizes shifted window partitions to enhance cross-window feature interaction. Specifically, the input to the left branch comprises feature maps extracted from both visible and thermal modalities, whereas the right branch processes cross-attention-enhanced visible features alongside the original thermal features. Two linear projection layers are subsequently applied separately: one to generate query vectors for the thermal image, and the other to generate keys and values for the visible features

$$Q = \text{Linear}\left(l_I\right) \tag{16}$$

$$(K, V) = \text{Linear}\left(l_V\right) \tag{17}$$

where l_I is the flattened sequence of thermal features f_I^a, and l_V is the flattened sequence of visible features f_V^a.

The generated queries, keys, and values undergo processing via multihead crossed attention (MHCA) mechanisms to conduct interactive attention computations between the visible and thermal modalities. This process enables the extraction of pertinent information across modalities, thereby enhancing the feature correlation between the two modalities and facilitating their subsequent integration

$$\text{Attention}\left(Q, K, V\right) = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{18}$$

$$Head_i = \text{Attention}\left(QW_{Qi}, KW_{Ki}, VW_{Vi}\right) \tag{19}$$

$$\text{MHCA}\left(Q, K, V\right) = \text{Cat}\left(\text{Head}_1, \ldots, \text{Head}_h\right)W_o \tag{20}$$

where d_k is the scaling factor, W_{Qi}, W_{Ki}, and W_{Vi} are the linear transformation matrices for each head, and W_O is the final linear transformation matrix.

The optimized features derived from the left branch, along with the original thermal features, are partitioned in the right branch via a shift window mechanism. Ultimately, the visible features f^F of the reconstructed image are extracted.

As shown in Fig. 1, after FRN optimization, the output features f^F are combined with f_V^D through residual connections to preserve the original modal information. The resulting feature is weighted by w_v, while the f_I^D is weighted by w_i, both of which are generated by the DFWG module. These two weighted feature representations are, then, concatenated and passed into the final detection head.

D. Loss Function

To ensure the effectiveness of single-mode feature extraction when integrated with the DSJE module, we employ a dual-supervision strategy by introducing a predetection head after DSJE. Notably, the predetection head is only active during training to constrain model optimization and does not participate in the inference phase. Consequently, the total loss function of IM-CMDet consists of two components: the detection head loss and the predetection head loss, both of which are computed using the same methodology. The loss functions utilized in this approach are defined as follows:

$$L_{\text{tot}} = \alpha L_{\text{det}} + (1 - \alpha)\, L_{\text{pre}} \tag{21}$$

$$L_{\text{det}} = L_{\text{reg}} + L_{\text{cls}} \tag{22}$$

$$L_{\text{pre}} = L_{p\text{reg}} + L_{p\text{cls}} \tag{23}$$

where L_{det} is the detection head loss, L_{pre} is the PreHead loss, and α represents the weight coefficient. $L_{\text{reg}}/L_{p\text{reg}}$ denote the bounding box regression loss in detection head and PreHead, respectively, and $L_{\text{cls}}/L_{p\text{cls}}$ denote the classification loss in detection head and PreHead, respectively.

The regression task is supervised by generalized intersection over union (GIoU) loss [54], and the classification task is supervised by focal loss [55]. The calculation methods for $L_{p\text{reg}}$ and $L_{p\text{cls}}$ are as follows:

$$L_{p\text{reg}} = \text{GIoU}\left(\text{box}_t, \text{box}^*\right) + \text{GIoU}\left(\text{box}_v, \text{box}^*\right) \tag{24}$$

$$L_{p\text{cls}} = \text{FL}\left(\text{cls}_t, \text{cls}^*\right) + \text{FL}\left(\text{cls}_v, \text{cls}^*\right) \tag{25}$$

where GIoU(·) and FL(·) represent the GIoU loss and focal loss, respectively. box and cls correspond to the regression and classification predictions, respectively. Subscripts t and v indicate the thermal branch and visible branch, respectively, while the superscript * denotes the object.

III. Experiment and Analysis

A. Datasets

The primary experiment in this study was conducted using the RGBTDronePerson [38] dataset. To further evaluate the generalization of the proposed method, we conducted additional tests on the VTUAV-det [38] and RTDOD [56] datasets.

1) RGBTDronePerson Dataset: The primary experiment in this study was conducted using the RGBTDronePerson [38] dataset, which comprises 6125 pairs of RGB and thermal images captured in a dual camera-equipped drone, DJI Mavia Enterprise Advanced, and annotated with a total of 70 880 object instances. The dataset comprehensively considers two primary factors: illumination and temperature. It encompasses both daytime and nighttime scenes, spanning a wide range of temperature conditions from summer to winter. The majority of images are captured at altitudes between 50 and 80 m, with pitch angles ranging from 45° to 60°, while a limited subset is obtained from vertical perspectives. For the purpose of experimentation, the RGBTDronePerson dataset has been partitioned into 4900 images for training and 1225 images for testing. Following TinyPerson [57], we use average precision (AP) as the performance metric, with intersection over union (IoU) thresholds of 0.25 and 0.50 defining true positive (TP) predictions.

2) VTUAV-det Dataset: The VTUAV-det dataset, reannotated by Zhang et al. [38], is derived from the extensive visible and infrared tracking benchmark VTUAV [58], covering a wide range of scenarios and weather conditions. The VTUAV dataset is captured by the DJI Matrice 300 RTK UAV with Zenmuse H20T camera. The thermal camera captures wavelengths in the 8–14 μm range. The flight height is controlled from 5 to 20 m. After recollection, the VTUAV-det dataset comprises 16 770 pairs of RGB and thermal images, with 11 392 pairs designated for training and 5378 pairs reserved for testing. These images contain annotations for 124 869 human instances, exhibiting significant size variations. Due

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TABLE I
PERFORMANCE COMPARISONS ON THE RGBTDRONEPERSON DATASET. THE BEST AND SECOND-BEST PERFORMANCES ARE BOLDED AND UNDERLINED, RESPECTIVELY

Method	mAP^{50}_{all}	mAP_{tiny50}	mAP^{50}_{tiny1}	mAP^{50}_{tiny2}	mAP^{50}_{tiny3}	mAP^{50}_{small}	mAP^{25}_{all}	FPS
Faster-rcnn	29.52	29.78	0.0	23.90	36.82	**37.73**	39.83	29.9
Retinanet	26.35	27.39	15.63	18.62	38.65	18.74	42.97	**33.2**
Cascade-rcnn	32.04	32.65	11.55	29.75	38.83	<u>33.18</u>	43.53	24.7
ATSS	36.45	37.50	17.39	24.85	43.16	28.15	51.28	31.0
ATSS_QLS	38.91	40.31	18.30	27.20	45.83	27.04	55.45	30.2
FCOS	29.72	30.67	15.45	24.54	33.89	27.52	44.46	<u>31.3</u>
HRFuser	28.11	28.67	2.24	31.28	33.63	26.40	38.43	11.4
TINet	26.03	26.38	0.0	20.20	32.43	29.72	36.95	24.4
CDC-Yolofusion	39.41	40.78	17.51	30.20	46.12	26.99	55.81	22.6
QFDet	<u>42.20</u>	<u>44.30</u>	**20.84**	30.90	<u>50.55</u>	26.23	57.65	21.4
C2Former	41.85	43.41	16.10	<u>31.76</u>	49.05	28.02	<u>58.01</u>	19.8
DPAL	39.35	40.82	18.17	29.25	46.91	25.20	55.61	9.7
IM-CMDet(Ours)	**43.61**	**45.18**	<u>18.92</u>	**32.09**	**51.37**	28.61	**58.81**	19.2

to the differences in collection conditions compared to the RGBTDronePerson dataset, the object sizes in VTUAV-det are relatively larger. Therefore, we adopt the mean AP (mAP) metric, as used in COCO [59], for evaluation. According to the COCO evaluation protocol, object sizes are categorized as "small," "medium," and "large." Specifically, the VTUAV-det training set includes 7651 small objects, 50 794 medium objects, and 17 869 large objects, while the testing set contains 9864 small objects, 32 731 medium objects, and 5960 large objects.

3) RTDOD Dataset: RTDOD [56] is a comprehensive large-scale dataset for visible RGB and thermal object detection, acquired using the DJI Matrice 300 RTK UAV with Zenmuse H20T camera. The dataset covers various scenarios such as suburbs, highways, bridges, as well as different weather conditions including sunny, rainy, and foggy, and a range of lighting conditions including day and night. It encompasses ten categories, including people, vehicles, trucks, and motorcycles. To address the significant category imbalance that could potentially compromise model performance, we refined the original RTDOD dataset by retaining only four categories: "Person," "Truck," "Motorcycle," and "Car." These categories collectively account for 94.71% of all instances. The refined dataset comprises 9578 pairs of training images and 3328 pairs of test images, totaling 169 446 instances. Importantly, the original train–test split was preserved after this filtering process. For both the VTUAV-det and RTDOD datasets, we adhere to the COCO [59] evaluation criteria.

B. Training and Testing Details

We implemented this model using PyTorch [60] and the MMDetectionV2 framework [61]. All image inputs were resized to 640 × 512 while maintaining their aspect ratio. The network backbone is ResNet50 [62], which was pretrained on ImageNet [63]. The model was trained for 12 epochs using the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, a learning rate of 0.01, a weight decay factor of 0.0001, and a batch size of 8. To prevent overfitting, we selected the epoch with the best validation performance as the final model. For data augmentation, we employed random flipping. During testing, the non-maximum suppression (NMS) IoU threshold for RGBTDronePerson was set to 0.35, while for VTUAV-det and RTDOD, it was set to 0.65. The confidence threshold was set to 0.05. All experiments were conducted on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory.

C. Comparison With State-of-the-Art Methods

In this section, we conduct a comprehensive comparison of IM-CMDet against several state-of-the-art object detection algorithms. These algorithms are categorized based on their approaches to small object detection and multimodal sensor fusion. For small object detection, we focus on two prominent methods: QFDet [38] and the adaptive training sample selection_quality-aware learning strategy (ATSS_QLS) detector [38], an enhanced variant of the ATSS detector [64]. Both methods have demonstrated significant improvements in performance, particularly for small objects, with the latter serving as a key baseline for our proposed method. For multimodal object detection, we compare IM-CMDet with leading sensor-fusion algorithms such as TINet [31], HRFuser [65], C2Former [42], CDC-YoloFusion [51], and DPAL [66]. Additionally, we evaluate baseline detectors for RGBT object detection, including Faster R-CNN [67], Cascade R-CNN [68], RetinaNet [55], FCOS [69], and ATSS [64]. These methods employ a two-stream architecture to extract features from visible and thermal modalities separately, followed by feature fusion through a 1 × 1 convolutional layer.

1) Results on RGBTDronePerson: We compared our method with the comparison methods on the RGBTDronePerson dataset in the same settings. The results are shown in Table I. "tiny1," "tiny2," "tiny3," and "small" denote absolute object sizes that are in the range of [0, 8), [9, 12), [12, 20), and [20, 32], respectively. From an overall performance perspective, IM-CMDet achieves an mAP^{all}_{50} of 43.61% and an mAP^{tiny}_{50} of 45.18%; on this dataset, we achieved optimal levels. In particular, IM-CMDet surpasses the baseline network by 4.7% and surpasses QFDet by 1.41%, which may be attributed to our object feature enhancement and interaction strategy for small objects. However, from the perspective of the tiny1 metric, QFDet outperforms our method, likely due to its quality-aware strategy based on point quality in the network.

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TABLE II

PERFORMANCE COMPARISONS ON THE VTUAV-DET DATASET. THE BEST AND SECOND-BEST PERFORMANCES ARE BOLDED AND UNDERLINED, RESPECTIVELY

Method	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l	FPS
Faster-rcnn	26.50	58.30	20.50	1.20	28.50	54.50	30.6
Retinanet	22.20	52.20	16.00	2.70	22.10	55.90	**33.0**
Cascade-rcnn	28.40	62.20	21.90	3.0	<u>30.40</u>	56.60	26.1
ATSS	28.50	66.00	20.30	8.4	28.20	55.60	31.3
ATSS_QLS	<u>30.70</u>	69.60	<u>22.60</u>	<u>12.40</u>	30.20	<u>56.90</u>	30.4
FCOS	27.40	65.10	19.10	7.5	27.50	53.00	<u>32.0</u>
HRFuser	28.50	61.80	22.10	4.40	30.40	55.20	11.5
TINet	26.70	60.20	19.70	1.90	29.30	51.50	26.4
CDC-YoloFusion	30.10	69.30	21.60	11.50	29.30	55.70	23.0
QFDet	30.60	<u>70.20</u>	22.00	12.20	30.10	<u>56.90</u>	22.7
C2Former	29.80	68.70	21.60	11.00	29.50	55.90	20.0
DPAL	29.60	68.40	21.50	11.00	29.10	56.20	9.7
IM-CMDet(Ours)	**31.50**	**70.70**	**23.00**	**12.80**	**30.70**	**57.80**	19.3

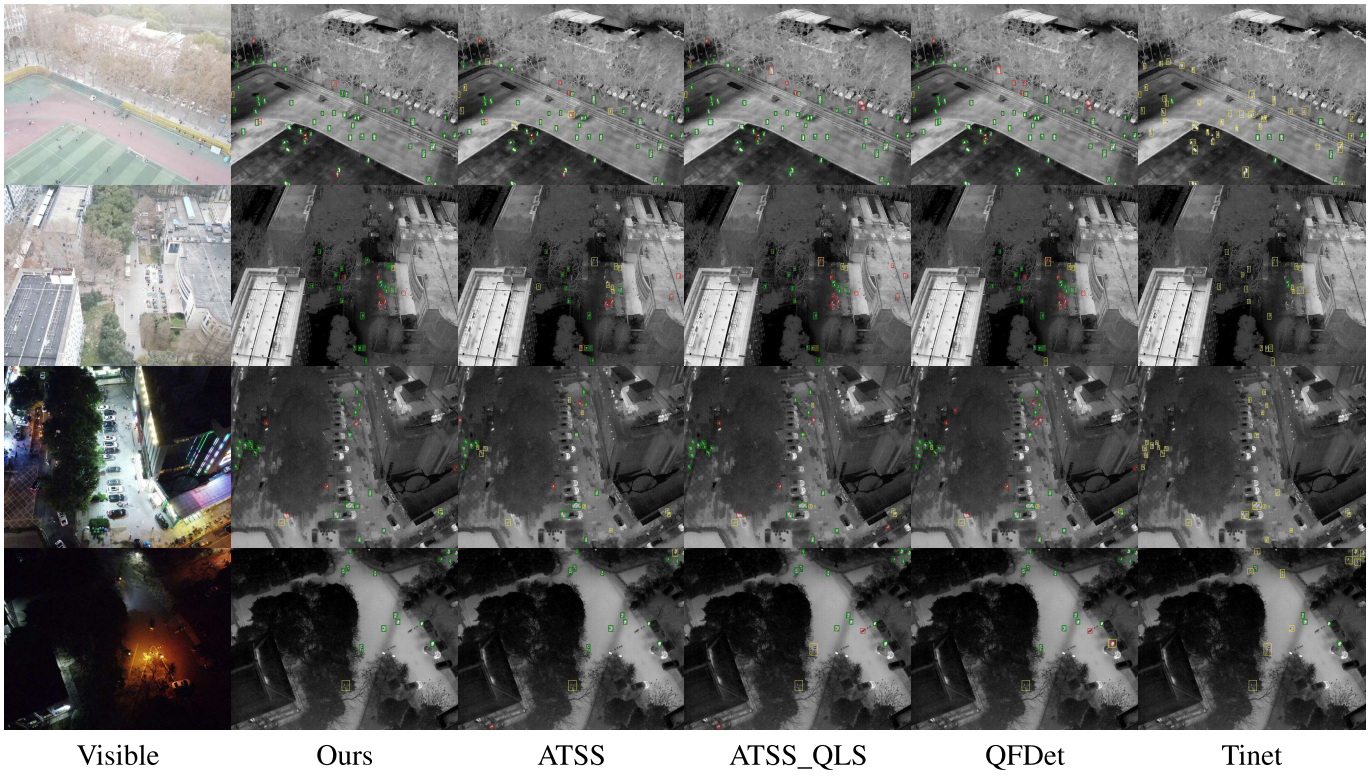


Fig. 5. Visualization of the selected methods on RGBTDronePerson.

Although our method performs well overall, its performance is lower compared to other methods on the small scale. It may be because objects at this scale only account for 2% of the dataset, which selectively diminishes learning for objects at this scale. C2Former [42] demonstrates strong performance on Tiny2 and mAP^{25}_{all} metrics, as previously shown in Table I. Nevertheless, these gains are accompanied by a substantial computational burden due to the extensive use of attention mechanisms, which limits its applicability in resource-constrained scenarios.

The qualitative comparisons among IM-CMDet, QFDet, ATSS, ATSS_QLS, and TINet are shown in Fig. 5. In the figure, the first column presents the original images from the visible modality, while the remaining columns display object detection results produced by various methods. The experimental results across various scenes demonstrate that, compared with the baseline approaches, the proposed method achieves a higher TP rate while maintaining a lower false positive (FP) rate. Specifically, in Scene 1, where the number of objects is relatively high, the proposed method achieves a more comprehensive detection performance. In Scene 3, the superior detection capability of the proposed method, particularly in cases where objects are visually similar to the background, can likely be attributed to the DSJE module, which enhances the features of small objects. Furthermore, with the assistance of the CSOFS module, specifically designed for small object detection, the network effectively integrates information from both modalities and improves its sensitivity to small objects. This results in significantly enhanced performance in Scenes 2 and 4. However, in Scenario 4, the model is unable to reliably detect targets situated in low-light conditions and heavily

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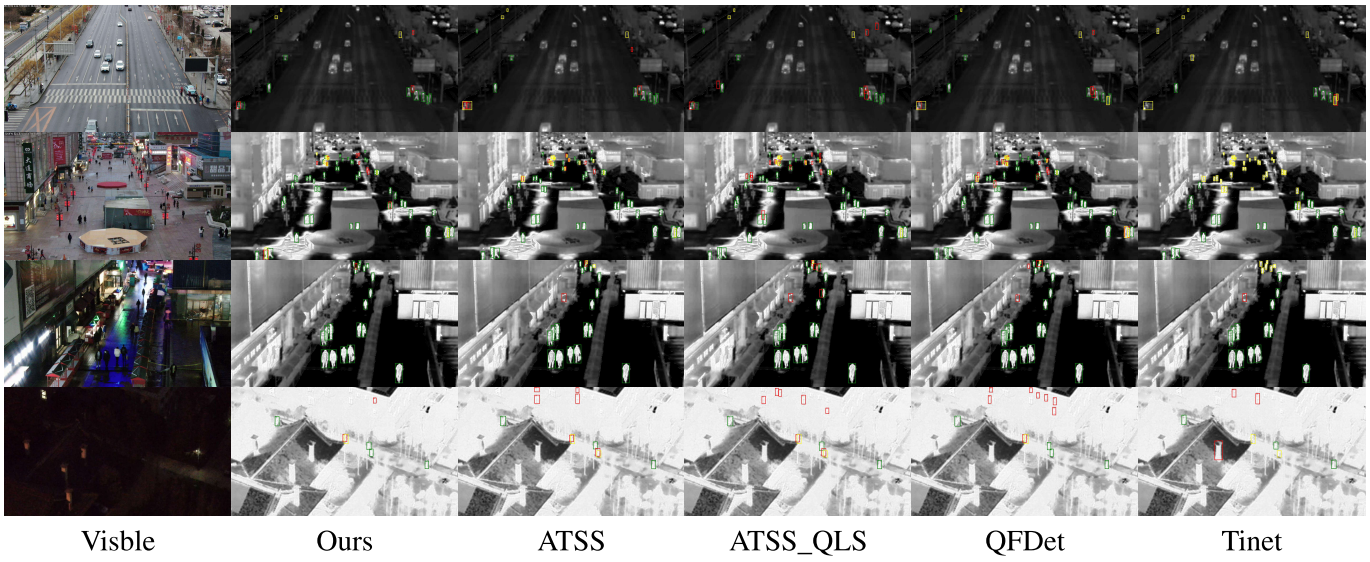


Fig. 6. Visualization of selected methods on VTUAV-det.

occluded by trees, where limited illumination and severe occlusion substantially weaken the visibility of discriminative features, leading to missed or uncertain detections.

In summary, the proposed method can more effectively utilize information from both RGB and thermal images, thereby better addressing the significant performance degradation caused by severe illumination variations in RGB images. Moreover, the examination of the third and fourth columns suggests that the modified ATSS_QLS algorithm achieves overall performance improvements over the original ATSS algorithm. Specifically, the ATSS_QLS algorithm demonstrates a higher TP rate, accompanied by fewer false and missed detections. These results indicate that the quality-aware learning strategy (QLS) assignment strategy effectively contributes to improved detection performance for small objects.

2) Results on VTUAV-det: On this dataset, the objects exhibit greater size variation, and we use the COCO standard to measure the performance of the comparison methods. The quantitative experimental results are shown in Table II. Compared to the ATSS_QLS method, our approach improves by 0.9% in mAP and mAP_{50} is 0.5% higher than the second-best method QFDet. In the mAP_s metric, our method is 0.6% higher than QFDet and 0.4% higher than ATSS_QLS, while in the mAP_l metric, it is 0.9% higher than QFDet. This indicates that our method also performs well under conditions of significant size variation, which may be attributed to the effectiveness of our detail-semantic joint enhancement module aimed at enhancing intramodal objects.

The subjective comparison results are shown in Fig. 6. Four scenes with varying lighting conditions were selected for testing to comprehensively evaluate the performance of the proposed method. Compared to other methods, our approach achieves a higher number of TP and fewer FP. Specifically, in Scene 1, where objects are unevenly distributed and clustered, the proposed detail enhancement strategy helps the network distinguish between objects, enabling more accurate localization. However, some missed detections still occur, which may be due to the model placing relatively more emphasis on the infrared modality, causing targets that do not exhibit distinctive features in the infrared image to be overlooked. In Scene 2, where small objects are densely distributed and the distinction between objects and the background is minimal, leading to poor object identifiability in the RGB modality, the combination of the detail enhancement method and the small-object feature interaction strategy further improves the network's sensitivity to objects, reducing missed and false detections. In Scenes 3 and 4, where lighting conditions are poor, the proposed modality interaction strategy allows our method to more effectively distinguish between similar objects in thermal images (e.g., in Scene 3, where a bright spot in the thermal image near a street-side shop is mistakenly identified as an object by comparative methods), achieving higher detection accuracy with fewer FP.

3) Results on RTDOD: Based on Table III, we compare the performance of various methods on the RTDOD dataset, emphasizing their detection capabilities across different object sizes. Our proposed method achieves the highest overall performance, with an mAP of 52.40%, an mAP_{50} of 81.40%, and an mAP_{75} of 58.60%. This improvement is primarily attributed to a substantial gain in small object detection (mAP_s: 37.10%), where our method significantly outperforms all other approaches. For medium- and large-scale objects, Cascade R-CNN exhibits strong performance, achieving the highest mAP_m (60.90%) and mAP_l (72.50%). Meanwhile, HRFuser, a cross-modal joint detection method, also shows competitive results, particularly in medium-sized object detection (mAP_m: 59.50%). However, both methods struggle with small object detection, which significantly limits their overall performance. These results highlight the advantage of our method in addressing the challenges of small object detection while maintaining competitive performance on other object scales.

Fig. 7 presents a qualitative comparison of different methods on the RTDOD dataset, highlighting the superior performance of our approach across diverse scenarios. In small-object detection tasks, particularly evident in the first row, our method accurately identifies distant and challenging objects that are often missed or incompletely detected by other methods such as ATSS and QFDet. Nevertheless, the proposed method fails to reliably detect camouflaged target that exhibits

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TABLE III

PERFORMANCE COMPARISONS ON THE RTDOD DATASET. THE BEST AND SECOND-BEST PERFORMANCES ARE BOLDED AND UNDERLINED, RESPECTIVELY

Method	mAP	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l	FPS
Faster-rcnn	47.70	72.40	54.40	24.10	59.20	71.70	31.4
Retinanet	39.40	68.10	40.20	15.10	48.00	71.10	**34.5**
Cascade-rcnn	49.80	73.80	<u>56.60</u>	27.30	**60.90**	<u>72.50</u>	26.8
ATSS	49.10	78.80	54.00	32.60	56.70	70.40	32.2
ATSS_QLS	<u>49.60</u>	79.80	54.60	33.40	56.80	70.40	31.6
FCOS	45.90	77.20	48.50	27.90	53.10	70.10	<u>32.7</u>
HRFuser	<u>49.60</u>	72.80	56.20	31.00	<u>59.50</u>	66.80	11.4
TINet	46.80	71.60	53.50	23.40	58.40	70.70	26.3
CDC-YoloFusion	49.00	79.10	53.70	<u>34.30</u>	56.20	68.40	23.1
QFDet	49.20	<u>81.10</u>	53.10	33.60	56.00	69.10	22.5
C2Former	48.60	79.20	52.80	34.00	55.30	67.70	20.6
DPAL	47.60	78.70	51.60	31.70	54.90	66.90	9.9
IM-CMDet(Ours)	**52.40**	**81.40**	**58.60**	**37.10**	59.10	**72.70**	19.5

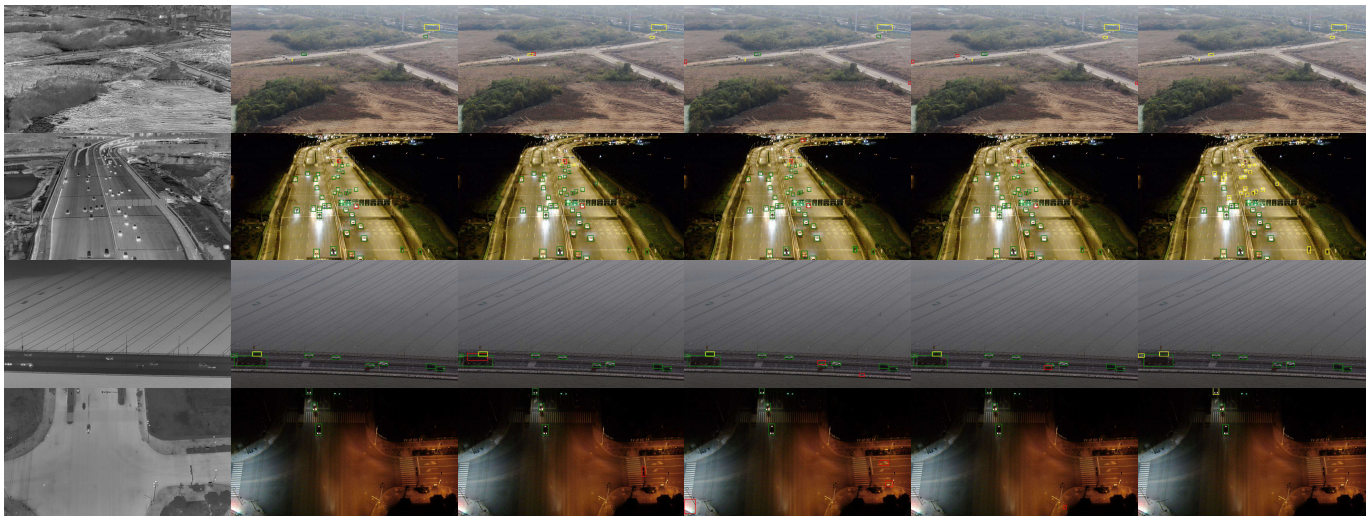


Fig. 7. Visualization of selected methods on RTDOD.

extremely low observability in both visible and infrared modalities within this scenario, which suggests that its scene interpretation capability could be further improved. In dense-object scenarios, as shown in the second row, our approach demonstrates precise localization and reduced false negatives, effectively balancing precision and recall in crowded scenes. Under foggy conditions depicted in the third row, our method exhibits remarkable robustness by accurately detecting objects obscured by reduced visibility, outperforming other methods like TINet. In the fourth row, the effectiveness of our method is further demonstrated by leveraging complementary features from both thermal and visible images, where the information from thermal images effectively filters out distracting features in the visible images, resulting in precise and reliable detection. These results underscore the advantages of our approach in cross-modal small-object detection tasks.

4) Efficiency Comparison: The backbones, parameter counts, and computational costs of the compared methods in this article are summarized in Table IV. To ensure a fair comparison, all comparison methods employ ResNet50 [62] as the backbone network. The only exception is HRFuser [65], where we keep the original backbone architecture in our experiments since its primary innovations lie in the backbone design itself. From Table IV, it can be observed that most recently proposed methods introduce a noticeable increase in computational cost

TABLE IV

COMPARISON OF NETWORK BACKBONES, PARAMETER COUNTS, AND COMPUTATIONAL COSTS FOR THE METHODS IN OUR EXPERIMENTS

Method	Backbone	Param.(MB)	GFLOPs
Faster-rcnn	ResNet50	97.17	111.77
Retinanet	ResNet50	59.82	98.97
Cascade-rcnn	ResNet50	92.61	113.8
ATSS	ResNet50	<u>59.48</u>	97.45
ATSS_QLS	ResNet50	<u>59.48</u>	97.45
FCOS	ResNet50	78.57	<u>94.41</u>
HRFuser	HRFormer	**48.79**	**52.92**
TINet	ResNet50	88.78	108.23
CDC-Yolofusion	ResNet50	83.54	98.09
QFDet	ResNet50	60.18	162.86
C2Former	ResNet50	118.69	145.54
DPAL	ResNet50	114.86	144.63
IM-CMDet(Ours)	ResNet50	105.04	133.93

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TABLE V
OVERALL ABLATION EXPERIMENTS. THE BEST PERFORMANCES ARE BOLDED

Method	DSJE	CSOFS	PreHead	mAP^{50}_{all}	Param.(MB)	GFlops	FPS
Baseline				38.91	**59.48**	**97.45**	**30.2**
M1	✓			40.33	76.36	102.75	25.3
M2		✓		40.22	88.12	125.81	21.5
M3			✓	41.27	59.48	97.45	29.9
M4	✓	✓		42.63	105.04	133.93	19.1
M5	✓		✓	41.90	76.36	102.75	24.7
M6		✓	✓	41.74	88.12	125.81	21.4
Ours	✓	✓	✓	**43.61**	105.04	133.93	19.2

TABLE VI
MODULEWISE PARAMETERS AND FLOPS

Module	DSJE	DFWG	FRN
Param.(MB)	12.3	17.7	10.9
Gflops	7.85	11.34	18.95

TABLE VII
ABLATION ON DSJE MODULE. THE BEST PERFORMANCES ARE BOLDED

Method	Detail Info	Enhancement Refe	mAP^{50}_{all}
M6			41.74
M7	✓		40.09
M8		✓	43.26
Ours	✓	✓	**43.61**

compared with the baseline. Although QFDet employs fewer parameters, it suffers from higher inference overhead due to the use of a computationally intensive predetection head during testing. In contrast, our method restricts this component to the training phase as supervision, thereby reducing inference complexity. Overall, while the additional modules in our approach lead to a moderate increase in parameter count, the model preserves competitive inference speed and incurs only a modest rise in GFlops, striking a favorable balance between efficiency and performance. A more detailed analysis of parameter size and computational cost for each component of the proposed framework is provided in Section III-D.

D. Ablation Study

1) Overall Ablation: Ablation studies were conducted on the RGBTDronePerson dataset to systematically evaluate the effectiveness of the proposed components. The results in Table V demonstrate that the proposed modules, fusion strategies, and training schemes are indeed beneficial. Specifically, the proposed method exhibited a 4.7% improvement in performance compared with the baseline. The incorporation of the DSJE module contributed a 1.42% performance enhancement to the network. The introduction of the fusion strategy yielded a 1.31% performance gain. Moreover, the three modules exhibit interdependent relationships and achieve collective enhancement—when operating individually, their effectiveness was somewhat constrained. Specifically, integrating DSJE modules with CSOFS methods resulted in further performance improvements, attributed to the synergistic combination of enhanced discriminative information and effective cross-modal fusion. By incorporating the dual-supervision scheme, the network achieves additional gains, highlighting the importance of enhancing feature extraction to improve fusion effectiveness. Notably, since the dual supervision is applied only during training, it introduces no additional parameters or computational cost during inference. Furthermore, its effects become more pronounced when jointly employed with the DSJE modules, where the enhanced supervision strategy further boosts the operational effectiveness of DSJE. Beyond performance evaluation, we also analyze model complexity in terms of parameter count, floating-point operations (FLOPs) and inference speed, as shown in Table V. Although the proposed method introduces a certain computational overhead, the performance gains—especially for small object detection—demonstrate a favorable tradeoff. To better illustrate the efficiency of each component, we provide a detailed breakdown of the parameter count and FLOPs for each module, as shown in Table VI. In particular, the DFWG module processes deep semantic features with high channel dimensions extracted from the backbone, resulting in a noticeable increase in parameter count while the associated growth in computational complexity remains relatively modest as the low spatial resolutions.

2) Ablation on DSJE: This study further examines the impact of the detail enhancement and ER components on overall performance in the DSJE module, as illustrated in Table VII. Compared with the M6 model, the integration of ER improves model performance by 1.52%, demonstrating that the hierarchical interaction strategy proposed in our method effectively optimizes multilevel feature representations. The elimination of detail enhancement from the original image processing pipeline results in a 0.35% performance decline, confirming that the proposed detail enhancement mechanism significantly enhances discriminative information of objects compared to the baseline network. However, removing the enhanced baseline module leads to a more severe degradation (1.65% performance drop) relative to the M6 model. One plausible explanation lies in that the isolated use of detail enhancement (without ER) introduces substantial noise from shallow high-frequency details extracted by the Laplacian filter. Specifically, shallow edge features often contain irrelevant structural information (e.g., vegetation or architectural

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TABLE VIII
ABLATION ON CSOFS MODULE. THE BEST PERFORMANCES ARE BOLDED

Method	DFWG	FRN	mAP^{50}_{all}
M5			41.90
M9	✓		42.25
M10		✓	42.18
Ours	✓	✓	**43.61**

edges) that can misguide network attention mechanisms. The joint implementation of ER and detail enhancement strategically amplifies object-specific features while suppressing background interference.

3) Ablation on CSOFS: In this study, we conducted systematic ablation experiments to evaluate the components of our proposed CSOFS, specifically designed for small-object detection. Comparative results against the M5 model reveal that both the DFWG and FRN modules contribute measurably to performance enhancement as summarized in Table VIII. While compared with the completed method, the removal of DFWG induces a marked performance deterioration (1.43% reduction), likely attributable to compromised redundancy filtering capacity and diminished sensitivity to small objects. Similarly, excluding FRN results in a 1.36% performance decline, potentially due to inadequate learning of cross-modal feature relevance compared to the complete configuration. This degradation suggests FRN's critical role in establishing intermodal semantic alignment, particularly for distinguishing subtle object characteristics from complex backgrounds (e.g., Scenarios 2 and 3 in Fig. 6).

To further validate the performance of the CSOFS strategy, we conducted comprehensive robustness tests by introducing artificially induced modal biases, as illustrated in Fig. 8. Specifically, two distinct perturbation strategies were employed: pixelwise translation and degree-by-degree center rotation, which were applied to both our method and the ablation variants. As demonstrated in Fig. 8(a), although the performance of all methods degrades under artificially introduced perturbations, our approach consistently maintains superior performance. The performance degradation analysis in Fig. 8(b) reveals that the incorporation of either FRN or DFWG significantly mitigated performance deterioration under different perturbation intensities when compared to the M5 method.

4) Visualization of Intermediate Feature: To further intuitively understand the roles of DSJE and FRN in our framework, we visualize the intermediate features, as shown in Fig. 9. From top to bottom, each column represents the original image, and the C3–C5 feature layers, respectively. It can be clearly observed that the features directly extracted by the backbone exhibit weak responses around small objects. After processing with the DSJE module, the object responses in both visible and infrared modalities are significantly enhanced, indicating that the hierarchical interaction and shallow detail injection strategies effectively improve feature detail preservation and enhance the representation capability across multiple scales. However, despite the

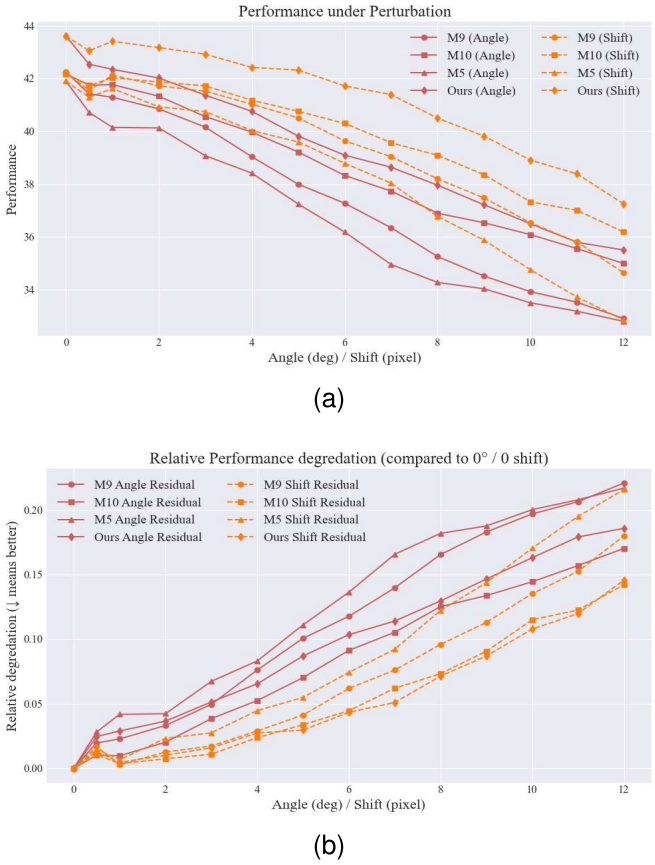


Fig. 8. Robustness analysis of four models under increasing perturbation intensity. (a) Performance under varying perturbations. (b) Relative degradation under varying perturbations.

improvements introduced by the DSJE module, inconsistencies in feature responses across modalities persist, and the visible modality still suffers from weakened representations under low illumination. A comparison between Fig. 9(d) and (e) reveals that the FRN module effectively compensates for the weak visible features by aligning them with infrared information. In addition, the response misalignment observed in Fig. 9(d) is noticeably alleviated after the application of the FRN module.

5) Sensitivity Analysis: The weighting configuration of loss functions critically governs network training preference. To identify the optimal weighting configuration for our dual supervision strategy, we conducted a systematic parameter sweep through progressive weight allocation. Specifically, we incremented the α coefficient from 0.1 to 0.9 in 0.1 intervals while maintaining a complementary β value ($\beta = 1 - \alpha$) to preserve weight summation unity. Our methodical evaluation reveals a distinct performance maximum at $\alpha = \beta = 0.5$, as quantitatively demonstrated in Fig. 10. This balanced configuration achieves an excellent balance between intramodal feature extraction and final detection results. Notably, performance degradation observed at extreme weight ratios ($\alpha < 0.3$ or $\alpha > 0.7$) underscores the necessity of balanced supervision—excessive emphasis on either component disrupts the delicate equilibrium required for effective small-object detection in complex environments.

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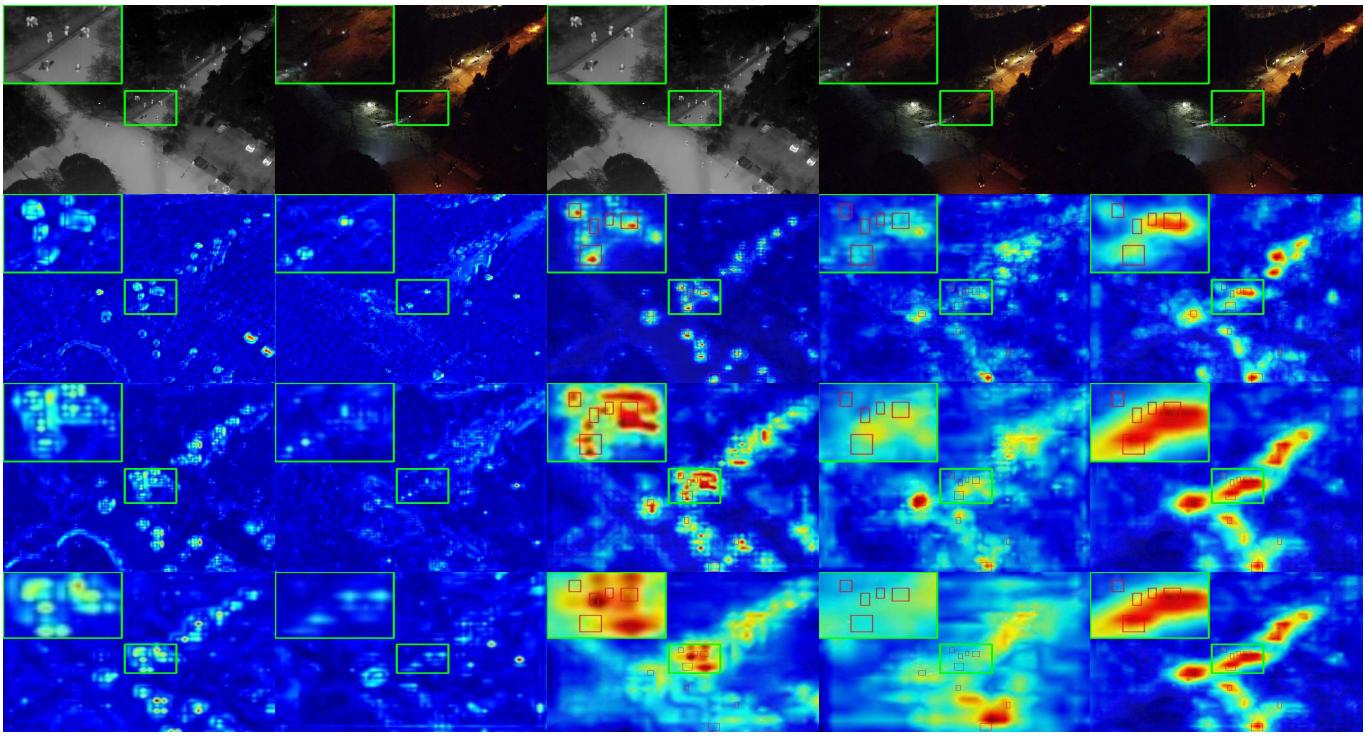


Fig. 9. Visualization of intermediate feature maps. (a) and (b) Infrared and visible features extracted by the backbone, respectively. (c) and (d) Corresponding features refined after DSJE module. (e) Visible features further processed through the FRN module.

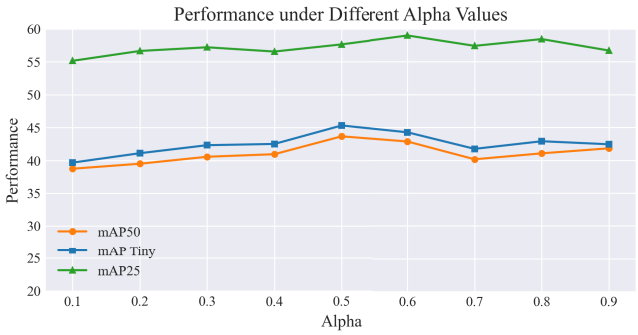


Fig. 10. Performance under different alpha values.

IV. Conclusion

In this article, we introduce a small object detection method based on the fusion of UAV aerial RGBT images, termed IM-CMDet. By integrating intramodel discriminative feature enhancement with a cross-modal feature alignment network specifically designed for small object characteristics, our method effectively enhances the performance of detecting small objects under varying illumination. Extensive experiments on three datasets demonstrate that our approach achieves superior performance in scenarios characterized by complex backgrounds.

Despite the significant advancements in accuracy and robustness of cross-modal fusion achieved by our approach, there remains potential for optimization implied in the results of experiments. In terms of inference efficiency, IM-CMDet demonstrates only sub-real-time performance, suggesting that there is still room for improving the efficiency of the model. However, in real-world scenarios, fast detection is crucial for better adaptation to video stream data. Future research could investigate more efficient attention mechanisms or adaptive window attention methods to enhance real-time detection performance. Moreover, the proposed fusion method may place excessive emphasis on the infrared modality under well-illuminated conditions, thereby attenuating the more discriminative features from the visible spectrum. Future work could incorporate illumination-aware mechanisms to dynamically balance multimodal feature representations, further enhancing overall detection performance.

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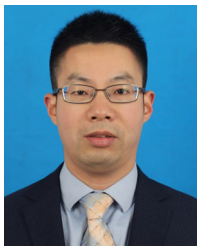
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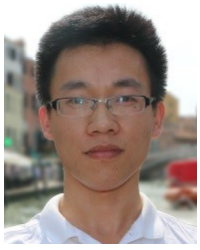
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